

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Experiment No. 2

Title:

Upload and analyze the data set given in CSV format and perform data preprocessing and visualization.

Aim:

To load a sales dataset (sales_data.csv) into a pandas DataFrame, clean and preprocess it (handling missing values), and analyze and visualize the data using bar plots, line plots, pivot tables, and a correlation heatmap.

Software Required:

Python 3, with the libraries pandas, numpy, matplotlib, and seaborn.

Description

The sample dataset sales_data.csv contains daily sales records with the columns Date, Product, Sales, Quantity, and Region. The dataset is loaded into a pandas DataFrame, checked for missing values, cleaned, and then summarized and visualized in multiple ways: a bar chart of total sales by product, a line chart of sales over time, a pivot table of sales by region and product, and a heatmap of the correlation between Sales and Quantity.

Sample Data Input (sales_data.csv)

```
Date,Product,Sales,Quantity,Region
2023-01-01,Product A,200,4,North
2023-01-02,Product B,150,3,South
2023-01-03,Product A,220,5,North
2023-01-04,Product C,300,6,East
2023-01-05,Product B,180,4,West
... (16 rows in total)
```

Program / Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the data into a pandas DataFrame
file_path = 'C:\\sales_data.csv'
df = pd.read_csv(file_path)

# Display the first few rows of the DataFrame
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
```

```

df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print(df.describe())

# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)

# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region',
                              columns='Product', aggfunc=np.sum, fill_value=0)

print(pivot_table)

# Correlation matrix
correlation_matrix = df[['Sales', 'Quantity']].corr()
print(correlation_matrix)

# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()

```

Output

1. First five rows of the dataset (df.head()):

	Date	Product	Sales	Quantity	Region
0	2023-01-01	Product A	200	4	North
1	2023-01-02	Product B	150	3	South
2	2023-01-03	Product A	220	5	North

3	2023-01-04	Product C	300	6	East
4	2023-01-05	Product B	180	4	West

2. Missing value check (df.isnull().sum()):

```
Date      0
Product    0
Sales      0
Quantity   0
Region     0
dtype: int64
```

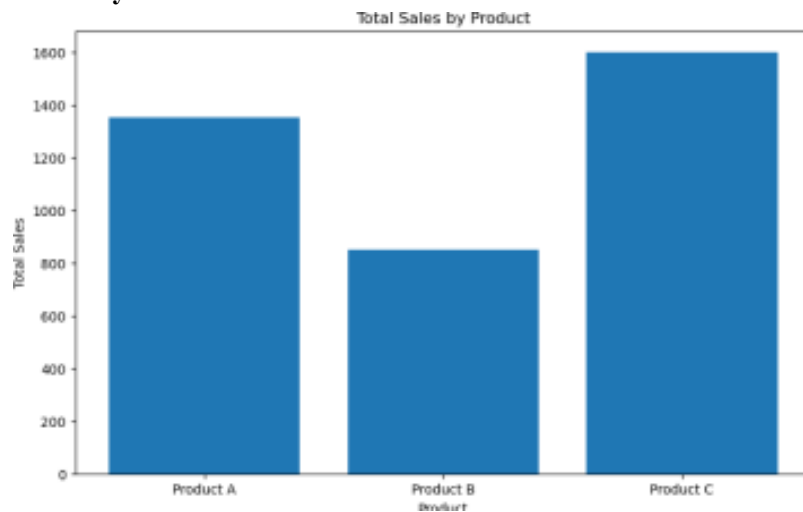
3. Summary statistics (df.describe()):

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

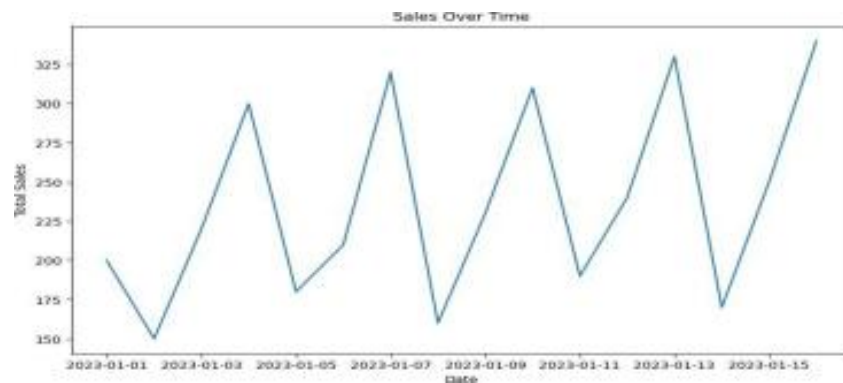
4. Product-wise sales and quantity summary:

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36

5. Bar plot – Total Sales by Product:



6. Line plot – Sales Over Time:



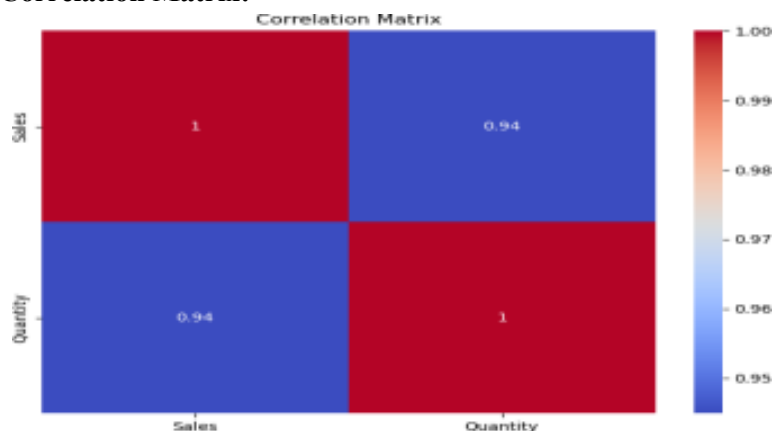
7. Pivot table – Sales by Region and Product:

Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

8. Correlation matrix:

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000

9. Heatmap of the Correlation Matrix:



Result

The given sales dataset was successfully loaded, checked for missing values, cleaned, and preprocessed using pandas. Total sales and quantity were summarized product-wise, and the data was visualized using a bar chart, a line chart, a pivot table, and a correlation heatmap. The correlation matrix shows a strong positive correlation (0.94) between Sales and Quantity, confirming that higher quantities sold are associated with higher sales revenue. Thus, the experiment on data preprocessing and visualization using pandas, matplotlib, and seaborn was executed successfully.